

CREATE A TABLE:-

DEPARTMENTS TABLE:-

```
CREATE TABLE Departments (  
    DepartmentID INT PRIMARY KEY,  
    DepartmentName VARCHAR(100) NOT NULL  
);
```

```
INSERT INTO Departments (DepartmentID, DepartmentName)  
VALUES  
(1, 'Computer Science'),  
(2, 'Mathematics');
```

[Show query box](#)

✓ MySQL returned an empty result set (i.e. zero rows). (Query took 0.0019 seconds.)

```
CREATE TABLE Departments ( DepartmentID INT PRIMARY KEY, DepartmentName VARCHAR(100) NOT NULL );
```

[\[Edit inline \]](#) [\[Edit \]](#) [\[Create PHP code \]](#)

✓ 2 rows inserted. (Query took 0.0006 seconds.)

```
INSERT INTO Departments (DepartmentID, DepartmentName) VALUES (1, 'Computer Science'), (2, 'Mathematics');
```

[\[Edit inline \]](#) [\[Edit \]](#) [\[Create PHP code \]](#)

STUDENTS TABLE:-

```
CREATE TABLE Students (  
    StudentID INT PRIMARY KEY,  
    FirstName VARCHAR(50) NOT NULL,  
    LastName VARCHAR(50) NOT NULL,  
    Email VARCHAR(100),  
    BirthDate DATE,  
    EnrollmentDate DATE  
);
```

```
INSERT INTO Students  
(StudentID, FirstName, LastName, Email, BirthDate, EnrollmentDate)  
VALUES  
(1, 'John', 'Doe', 'john@email.com', '2000-01-15', '2022-08-01'),  
(2, 'Jane', 'Smith', 'jane@email.com', '1999-05-25', '2021-08-01'),  
(3, 'Rahul', 'Mehta', 'rahul@email.com', '2001-03-10', '2023-01-15'),  
(4, 'Priya', 'Shah', 'priya@email.com', '2000-07-20', '2023-07-10'),  
(5, 'Neha', 'Patel', 'neha@email.com', '2002-02-18', '2024-01-12'),  
(6, 'Rohan', 'Desai', 'rohan@email.com', '1999-11-05', '2020-06-15'),
```

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```
(7, 'Aisha', 'Khan', 'aisha@email.com', '2001-09-14', '2024-08-20'),
(8, 'Karan', 'Joshi', 'karan@email.com', '2000-12-30', '2022-09-05'),
(9, 'Meera', 'Shah', 'meera@email.com', '2002-04-11', '2023-08-25'),
(10, 'Arjun', 'Patel', 'arjun@email.com', '1998-10-19', '2019-07-01'),
(11, 'Sneha', 'Mehta', 'sneha@email.com', '2001-01-22', '2024-02-10');
```

```
MySQL returned an empty result set (i.e. zero rows) (Query took 0.000 seconds)

CREATE TABLE Students ( StudentID INT PRIMARY KEY, FirstName VARCHAR(50) NOT NULL, LastName VARCHAR(50) NOT NULL, Email VARCHAR(100), BirthDate DATE, EnrollDate DATE );
Explain [[1.0]] Create PHP code

11 rows inserted. (Query took 0.000 seconds)

INSERT INTO Students (StudentID, FirstName, LastName, Email, BirthDate, EnrollDate) VALUES (1, 'Jen', 'Kim', 'johndoe@email.com', '2000-05-15', '2023-09-01'), (2, 'Jen', 'Smith', 'jensmith@email.com', '2000-05-25', '2023-09-01'), (3, 'Rahul', 'Mehta', 'rahulmehta@email.com', '2003-07-10', '2023-07-15'), (4, 'Priya', 'Sharma', 'priyasharma@email.com', '2000-07-20', '2023-07-25'), (5, 'Sara', 'Patel', 'sarapatel@email.com', '2003-03-18', '2023-03-23'), (6, 'Rahul', 'Doshi', 'rahuldoshi@email.com', '1999-11-05', '2020-09-10'), (7, 'Aisha', 'Khan', 'aishakhan@email.com', '2001-09-14', '2024-08-20'), (8, 'Karan', 'Joshi', 'karanjoshi@email.com', '2000-12-30', '2022-09-05'), (9, 'Meera', 'Shah', 'meerashah@email.com', '2002-04-11', '2023-08-25'), (10, 'Arjun', 'Patel', 'arjunpatel@email.com', '1998-10-19', '2019-07-01'), (11, 'Sneha', 'Mehta', 'snehamehta@email.com', '2001-01-22', '2024-02-10');
```

COURSES TABLE:-

```
CREATE TABLE Courses (
  CourseID INT PRIMARY KEY,
  CourseName VARCHAR(100) NOT NULL,
  DepartmentID INT,
  Credits INT,
  FOREIGN KEY (DepartmentID) REFERENCES Departments(DepartmentID)
);
```

```
INSERT INTO Courses
(CourseID, CourseName, DepartmentID, Credits)
VALUES
(101, 'Introduction to SQL', 1, 3),
(102, 'Data Structures', 2, 4),
(103, 'Database Management Systems', 1, 4),
(104, 'Statistics', 2, 3),
(105, 'Calculus', 2, 4);
```

```
MySQL returned an empty result set (i.e. zero rows) (Query took 0.000 seconds)

CREATE TABLE Courses ( CourseID INT PRIMARY KEY, CourseName VARCHAR(100) NOT NULL, DepartmentID INT, Credits INT, FOREIGN KEY (DepartmentID) REFERENCES Departments(DepartmentID) );
Explain [[1.0]] Create PHP code

5 rows inserted. (Query took 0.000 seconds)

INSERT INTO Courses (CourseID, CourseName, DepartmentID, Credits) VALUES (101, 'Introduction to SQL', 1, 3), (102, 'Data Structures', 2, 4), (103, 'Database Management Systems', 1, 4), (104, 'Statistics', 2, 3), (105, 'Calculus', 2, 4);
```

INSTRUCTORS TABLE:-

```
CREATE TABLE Instructors (
  InstructorID INT PRIMARY KEY,
  FirstName VARCHAR(50) NOT NULL,
  LastName VARCHAR(50) NOT NULL,
  Email VARCHAR(100),
  DepartmentID INT,
  Salary DECIMAL(10,2),
```

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FOREIGN KEY (DepartmentID) REFERENCES Departments(DepartmentID)
);

INSERT INTO Instructors

(InstructorID, FirstName, LastName, Email, DepartmentID, Salary)

VALUES

(1, 'Alice', 'Johnson', 'alice@univ.com', 1, 50000.00),

(2, 'Bob', 'Lee', 'bob@univ.com', 2, 55000.00),

(3, 'Carol', 'Brown', 'carol@univ.com', 1, 60000.00);

✓ MySQL returned an empty result set (i.e. zero rows). (Query took 0.005 seconds)

```
CREATE TABLE Instructors (InstructorID INT PRIMARY KEY, FirstName VARCHAR(50) NOT NULL, LastName VARCHAR(50) NOT NULL, Email VARCHAR(100), DepartmentID INT, Salary DECIMAL(10,2), FOREIGN KEY (DepartmentID) REFERENCES Departments(DepartmentID)) ;
```

[Job done !! 0:01] [Query took 0.00s]

✓ 3 rows inserted. (Query took 0.005 seconds)

```
INSERT INTO Instructors (InstructorID, FirstName, LastName, Email, DepartmentID, Salary) VALUES ((1, 'Alice', 'Johnson', 'alice@univ.com', 1, 50000.00), (2, 'Bob', 'Lee', 'bob@univ.com', 2, 55000.00), (3, 'Carol', 'Brown', 'carol@univ.com', 1, 60000.00);
```

ENROLLMENTS TABLE:-

CREATE TABLE Enrollments (

EnrollmentID INT PRIMARY KEY,

StudentID INT,

CourseID INT,

EnrollmentDate DATE,

FOREIGN KEY (StudentID) REFERENCES Students(StudentID),

FOREIGN KEY (CourseID) REFERENCES Courses(CourseID)

);

INSERT INTO Enrollments

(EnrollmentID, StudentID, CourseID, EnrollmentDate)

VALUES

(1, 1, 101, '2022-08-01'),

(2, 2, 101, '2021-08-01'),

(3, 3, 101, '2023-01-15'),

(4, 4, 101, '2023-07-10'),

(5, 5, 101, '2024-01-12'),

(6, 6, 101, '2020-06-15'),

(7, 7, 101, '2024-08-20'),

(8, 8, 101, '2022-09-05'),

(9, 9, 101, '2023-08-25'),

(10, 10, 101, '2019-07-01'),

(11, 11, 101, '2024-02-10'),

(12, 1, 102, '2022-08-05'),

(13, 2, 102, '2021-08-05'),

(14, 3, 102, '2023-01-20'),

(15, 4, 102, '2023-07-15');

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MySQL returned an empty result set (i.e. zero rows) (Query took 0.0004 seconds)

```
CREATE TABLE Students (
  StudentID INT PRIMARY KEY,
  CourseID INT,
  EnrollmentDate DATE,
  FOREIGN KEY (StudentID) REFERENCES Students(StudentID),
  FOREIGN KEY (CourseID) REFERENCES Courses(CourseID)
)
-- table [[15.6]] Create PHP code
```

15 rows inserted (Query took 0.0003 seconds)

```
INSERT INTO Departments (DepartmentID, DepartmentName, EnrollmentDate) VALUES
(1, 'Computer Science', '2022-09-01'),
(2, 'Mathematics', '2022-09-01'),
(3, 'Physics', '2022-09-01'),
(4, 'Biology', '2022-09-01'),
(5, 'Chemistry', '2022-09-01'),
(6, 'Geology', '2022-09-01'),
(7, 'Astronomy', '2022-09-01'),
(8, 'Environmental Science', '2022-09-01'),
(9, 'Earth Science', '2022-09-01'),
(10, 'Oceanography', '2022-09-01'),
(11, 'Atmospheric Science', '2022-09-01'),
(12, 'Planetary Science', '2022-09-01'),
(13, 'Cosmology', '2022-09-01'),
(14, 'Astrophysics', '2022-09-01'),
(15, 'Astrobiology', '2022-09-01')
-- table [[15.6]] Create PHP code
```

1. CURD OPERATION :-

ON DEPARTMENTS:-

- CREATE

INSERT INTO Departments (DepartmentID, DepartmentName) VALUES (3, 'Physics');

- READ

SELECT * FROM Departments;

- UPDATE

UPDATE Departments SET DepartmentName = 'Physics and Applied Science' WHERE DepartmentID = 3;

- DELETE

DELETE FROM Departments WHERE DepartmentID = 3;

SELECT * FROM Departments;

Showing rows 0 - 1 (2 total, Query took 0.0003 seconds.)

```
SELECT * FROM Departments;
```

Profiling [Edit inline] [Edit] [Explain SQL] [Create PHP code] [Refresh]

☐ Show all | Number of rows: 25 | Filter rows: Search this table | Sort by key: None

Extra options

DepartmentID DepartmentName

☐ Edit ☐ Copy ☐ Delete 1 Computer Science

☐ Edit ☐ Copy ☐ Delete 2 Mathematics

☐ Check all With selected: ☐ Edit ☐ Copy ☐ Delete ☐ Export

☐ Show all | Number of rows: 25 | Filter rows: Search this table | Sort by key: None

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ON STUDENTS :-

- CREATE

INSERT INTO Students (StudentID, FirstName, LastName, Email, BirthDate, EnrollmentDate) VALUES (12, 'Amit', 'Patel', 'amit@email.com', '2001-06-15', '2024-09-01');

- READ

SELECT * FROM Students;

- UPDATE

UPDATE Students SET Email = 'amit.patel@email.com' WHERE StudentID = 12;

- DELETE

DELETE FROM Students WHERE StudentID = 12;

SELECT * FROM Students;

	StudentID	FirstName	LastName	Email	BirthDate	EnrollmentDate
☐ Edit Copy Delete	1	John	Doe	john@email.com	2000-01-15	2022-08-01
☐ Edit Copy Delete	2	Jane	Smith	jane@email.com	1999-05-25	2021-08-01
☐ Edit Copy Delete	3	Rahul	Mehta	rahul@email.com	2001-03-10	2023-01-15
☐ Edit Copy Delete	4	Priya	Shah	priya@email.com	2000-07-20	2023-07-10
☐ Edit Copy Delete	5	Neha	Patel	neha@email.com	2002-02-18	2024-01-12
☐ Edit Copy Delete	6	Rohan	Desai	rohan@email.com	1999-11-05	2020-06-15
☐ Edit Copy Delete	7	Aisha	Khan	aisha@email.com	2001-09-14	2024-08-20
☐ Edit Copy Delete	8	Karan	Joshi	karan@email.com	2000-12-30	2022-09-05
☐ Edit Copy Delete	9	Meera	Shah	meera@email.com	2002-04-11	2023-08-25
☐ Edit Copy Delete	10	Arjun	Patel	arjun@email.com	1998-10-19	2019-07-01
☐ Edit Copy Delete	11	Sneha	Mehta	sneha@email.com	2001-01-22	2024-02-10
☐ Edit Copy Delete	12	Amit	Patel	amit@email.com	2001-06-15	2024-09-01

✓ 1 row affected. (Query took 0.0006 seconds.)

UPDATE Students SET Email = 'amit.patel@email.com' WHERE StudentID = 12;

[Edit inline] [Edit] [Create PHP code]

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Extra options

		StudentID	FirstName	LastName	Email	BirthDate	EnrollmentDate
☐	 Edit  Copy  Delete	1	John	Doe	john@email.com	2000-01-15	2022-08-01
☐	 Edit  Copy  Delete	2	Jane	Smith	jane@email.com	1999-05-25	2021-08-01
☐	 Edit  Copy  Delete	3	Rahul	Mehta	rahul@email.com	2001-03-10	2023-01-15
☐	 Edit  Copy  Delete	4	Priya	Shah	priya@email.com	2000-07-20	2023-07-10
☐	 Edit  Copy  Delete	5	Neha	Patel	neha@email.com	2002-02-18	2024-01-12
☐	 Edit  Copy  Delete	6	Rohan	Desai	rohan@email.com	1999-11-05	2020-06-15
☐	 Edit  Copy  Delete	7	Aisha	Khan	aisha@email.com	2001-09-14	2024-08-20
☐	 Edit  Copy  Delete	8	Karan	Joshi	karan@email.com	2000-12-30	2022-09-05
☐	 Edit  Copy  Delete	9	Meera	Shah	meera@email.com	2002-04-11	2023-08-25
☐	 Edit  Copy  Delete	10	Arjun	Patel	arjun@email.com	1998-10-19	2019-07-01
☐	 Edit  Copy  Delete	11	Sneha	Mehta	sneha@email.com	2001-01-22	2024-02-10

2. RETRIVE STUDENT WHO ENROOLED AFTER 2022.:-

```
SELECT *  
FROM Students  
WHERE EnrollmentDate > '2022-12-31';
```

Output

StudentID	FirstName	LastName	Email	BirthDate	EnrollmentDate
3	Rahul	Mehta	rahul@email.com	2001-03-10	2023-01-15
4	Priya	Shah	priya@email.com	2000-07-20	2023-07-10
5	Neha	Patel	neha@email.com	2002-02-18	2024-01-12
7	Aisha	Khan	aisha@email.com	2001-09-14	2024-08-20
9	Meera	Shah	meera@email.com	2002-04-11	2023-08-25
11	Sneha	Mehta	sneha@email.com	2001-01-22	2024-02-10

3. RETRIVE COURSE WITH LIMIT 5.:-

```
SELECT *  
FROM Courses  
WHERE DepartmentID = 2  
LIMIT 5;
```

Output

CourseID	CourseName	DepartmentID	Credits
102	Data Structures	2	4
104	Statistics	2	3
105	Calculus	2	4

4.FILTERING DATA:-

```
SELECT CourseID, COUNT(StudentID) AS StudentCount  
FROM Enrollments  
GROUP BY CourseID  
HAVING COUNT(StudentID) > 5;
```

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CourseID	StudentCount
101	11

5.FIND STUDENTS WHO ARE ENROLLED IN BOTH SQL&DATA STRUCTURE:-

```
SELECT StudentID
FROM Enrollments
WHERE CourseID IN (101, 102)
GROUP BY StudentID
HAVING COUNT(DISTINCT CourseID) = 2;
```

Output

StudentID
1
2
3
4

6. Find students who are either enrolled to SQL or Data STRUCTURE:-

```
SELECT DISTINCT
    s.StudentID,
    s.FirstName,
    s.LastName
FROM Students s
JOIN Enrollments e
ON s.StudentID = e.StudentID
WHERE e.CourseID IN (101, 102);
```

StudentID	FirstName	LastName
1	John	One
2	Jane	Smith
3	Rahul	Malhotra
4	Priya	Shah
5	Neha	Rajal
6	Rohan	Desai
7	Anshu	Khan
8	Ravi	Joshi
9	Meera	Shah
10	Arun	Rajal
11	Devika	Malhotra

7. Calculate the average number of credits for all courses.:-

```
SELECT AVG(Credits) AS AverageCredits
FROM Courses;
```

AverageCredits

3.6

8. Find the maximum salary:-

```
SELECT MAX(Salary) AS MaximumSalary
FROM Instructors
WHERE DepartmentID = 1;
```

MaximumSalary

60000

9. Count the number of students enrolled in each department.

```
SELECT d.DepartmentName,
       COUNT(DISTINCT e.StudentID) AS StudentCount
FROM Departments d
JOIN Courses c
ON d.DepartmentID = c.DepartmentID
JOIN Enrollments e
ON c.CourseID = e.CourseID
GROUP BY d.DepartmentID, d.DepartmentName;
```

Output:

DepartmentName	StudentCount
Computer Science	11
Mathematics	4

10. Perform INNER JOIN to retrieve students and their corresponding courses.:-

```
SELECT
    s.FirstName,
    s.LastName,
    c.CourseName
FROM Students s
INNER JOIN Enrollments e
ON s.StudentID = e.StudentID
INNER JOIN Courses c
ON e.CourseID = c.CourseID;
```

Output

FirstName	LastName	CourseName
John	Doe	Introduction to SQL
Jane	Smith	Introduction to SQL
Rahul	Mehta	Introduction to SQL
Priya	Shah	Introduction to SQL
Neha	Patel	Introduction to SQL
Rohan	Desai	Introduction to SQL
Aisha	Khan	Introduction to SQL
Karan	Joshi	Introduction to SQL
Meera	Shah	Introduction to SQL
Arjun	Patel	Introduction to SQL
Sneha	Mehta	Introduction to SQL
John	Doe	Data Structures
Jane	Smith	Data Structures
Rahul	Mehta	Data Structures

11. Perform LEFT JOIN:-

```
SELECT
    s.StudentID,
    s.FirstName,
    s.LastName,
    c.CourseName
FROM Students s
LEFT JOIN Enrollments e
ON s.StudentID = e.StudentID
LEFT JOIN Courses c
ON e.CourseID = c.CourseID;
```

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StudentID	FirstName	LastName	CourseName
1	John	Doe	Introduction to SQL
1	John	Doe	Data Structures
2	Jane	Smith	Introduction to SQL
2	Jane	Smith	Data Structures
3	Rahul	Mehta	Introduction to SQL
3	Rahul	Mehta	Data Structures
4	Priya	Shah	Introduction to SQL
4	Priya	Shah	Data Structures
5	Neha	Patel	Introduction to SQL
6	Rohan	Desai	Introduction to SQL
7	Aisha	Khan	Introduction to SQL
8	Karan	Joshi	Introduction to SQL
9	Meera	Shah	Introduction to SQL
10	Arjun	Patel	Introduction to SQL

12. Subquery: courses that have more than 10 students.:-

```

SELECT DISTINCT
    s.StudentID,
    s.FirstName,
    s.LastName
FROM Students s
WHERE s.StudentID IN
(
    SELECT e.StudentID
    FROM Enrollments e
    WHERE e.CourseID IN
    (
        SELECT CourseID
        FROM Enrollments
        GROUP BY CourseID
        HAVING COUNT(StudentID) > 10
    )
);

```

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Output

StudentID	FirstName	LastName
1	John	Doe
2	Jane	Smith
3	Rahul	Mehta
4	Priya	Shah
5	Neha	Patel
6	Rohan	Desai
7	Aisha	Khan
8	Karan	Joshi
9	Meera	Shah
10	Arjun	Patel
11	Sneha	Mehta

13. EXRRACT THE YEAR FROM THE ENROLLMENTDATE OF STUDENTS:-

SELECT StudentID, FirstName, LastName,
 strftime('%Y', EnrollmentDate) AS Enrollment_Year
FROM Students;

StudentID	FirstName	LastName	Enrollment_Year
1	John	Doe	2022
2	Jane	Smith	2021
3	Rahul	Mehta	2023
4	Priya	Shah	2023
5	Neha	Patel	2024
6	Rohan	Desai	2020
7	Aisha	Khan	2024
8	Karan	Joshi	2022
9	Meera	Shah	2023
10	Arjun	Patel	2019
11	Sneha	Mehta	2024

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14. Concatenate the instructor's first and last name.:-

```
SELECT CONCAT(FirstName, ' ', LastName) AS InstructorName
FROM Instructors;
```

15. Calculate the average number of students enrolled in courses.:-

```
SELECT AVG(StudentCount) AS AverageStudents
FROM
(
    SELECT CourseID, COUNT(StudentID) AS StudentCount
    FROM Enrollments
    GROUP BY CourseID
) AS CourseEnrollment;
```

16. Label students as "Senior" or "Junior" based on their year of enrollment.:-

```
SELECT
    StudentID,
    FirstName,
    LastName,
    CASE
        WHEN DATEDIFF(CURDATE(), EnrollmentDate) > 4 * 365
        THEN 'Senior'
        ELSE 'Junior'
    END AS StudentStatus
FROM Students;
```

StudentID	FirstName	LastName	EnrollmentDate	Student_Status
1	John	Doe	2022-08-01	Senior
2	Jane	Smith	2021-06-01	Senior
3	Rahul	Mehta	2023-01-15	Junior
4	Priya	Shah	2023-07-10	Junior
5	Neha	Patel	2024-01-12	Junior
6	Rohan	Desai	2020-06-15	Senior
7	Aisha	Khan	2024-08-20	Junior
8	Karan	Joshi	2022-09-05	Senior
9	Meera	Shah	2023-08-25	Junior

